

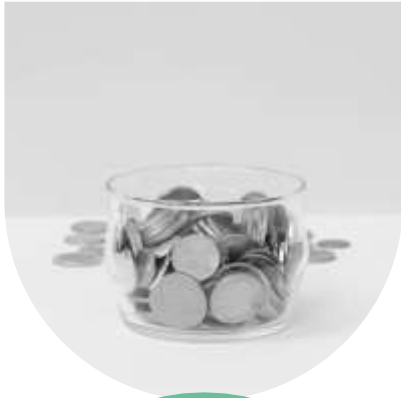
HOME LOAN APPROVAL ANALYSIS

Exploratory Data Analysis
with Python



Divyanshi Nigam

Home Loan



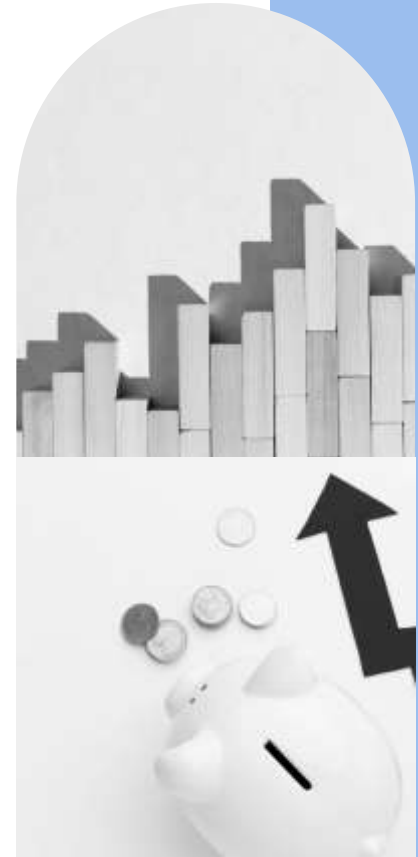
Owning a home is a lifelong dream for many and often represents one of the biggest financial decisions an individual makes.

A home loan is a secured financial product offered by banks or housing finance institutions, where the property itself acts as collateral.

These loans help individuals finance the purchase, construction, renovation, or extension of residential properties, with repayment structured through monthly installments over a fixed tenure.

—Introduction

The dataset is sourced from Skill Circle and contains valuable information for analysis. The primary focus of this assessment is on data exploration and visualization.



—Objective

- Gain familiarity with the dataset.
- Identify patterns, trends, and potential insights.
- Perform data exploration and visualization. Generate meaningful visualizations to communicate your findings.
- This project aims to predict loan approval based on the given dataset.
- It involves data cleaning, data analysis, pre-processing to gain beneficial derivatives.

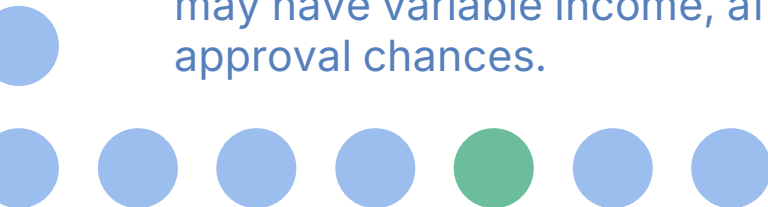


Data Dictionary

Feature	Type	Description
Loan_ID	Object	Unique identifier for each loan application.
Gender	Object	Gender of the applicant (Male/Female).
Married	Object	Marital status of the applicant (Yes/No).
Dependents	Object	Number of dependents economically reliant on the applicant (0, 1, 2, 3+).
Education	Object	Educational qualification of the applicant (Graduate/Not Graduate).
Self_Employed	Object	Indicates whether the applicant is self-employed (Yes/No).
ApplicantIncome	Int64	Monthly income of the applicant.
CoapplicantIncome	Int64	Monthly income of the co-applicant (if any).
LoanAmount	Float64	Total loan amount requested by the applicant.
Loan_Amount_Term	Float64	Duration of loan repayment in months (e.g., 360 months for 30 years).
Credit_History	Int64	Binary indicator of credit history (1 = Good credit, 0 = Bad credit).
Property_Area	Object	Geographical area where the property is located (Urban/Semi-urban/Rural).

Key Features

1. **Loan_ID:** A unique identifier assigned to each loan application for tracking and reference purposes.
2. **Gender :** The gender of the applicant (e.g., Male, Female).
3. **Married :** Marital status can influence financial obligations and risk assessment.
4. **Dependents :** This factor may impact loan eligibility.
5. **Education:** Higher education levels may correlate with better job prospects and higher income.
6. **Self_Employed:** Self-employed applicants may have variable income, affecting loan approval chances.



7. **ApplicantIncome:** This is a critical factor in determining loan eligibility and repayment capacity.
8. **CoapplicantIncome:** A higher combined income can increase the chances of loan approval.
9. **LoanAmount:** Loan amount is typically based on income, creditworthiness, and other factors.
10. **Loan_Amount_Term:** The duration (in months) over which the loan is to be repaid. Common values include 360 months (30 years), 240 months (20 years), etc.
11. **Credit_History**
12. A value of **1** suggests good repayment behavior, while **0** indicates a riskier credit profile. Some values may be missing.
13. **Property_Area:** location can impact its value and the loan approval process.



Step 1: Installation and Importing of Libraries

```
!pip install pandas
!pip install numpy
!pip install matplotlib
!pip install seaborn
```

```
55] #IMPORTING LIBRARIES
import os ## optional library - used to import paths for different files
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

import warnings
warnings.filterwarnings('ignore')
```

1. **numpy (np)**– Used for numerical computations and handling arrays efficiently
2. **pandas (pd)**– Provides data manipulation and analysis tools, mainly for working with DataFrames.
3. **seaborn (sns)**– Enhances data visualization by creating aesthetically pleasing statistical graphics.
4. **matplotlib.Pyplot**–A plotting library used for creating static, animated, and interactive visualization.

Step 2: Data Collection and Loading

I have used Google Colab as my development environment for this project, as it provides a cloud based Jupyter Notebook with powerful computing capabilities.

```
] # Load the dataset into a Python environment  
df=pd.read_csv('/content/drive/MyDrive/Colab Notebooks/EDA DATASETS/loan_sanction_test.csv')
```

Step 3: Basic Data Inspection

```
df.head()
```

	Loan_ID	Gender	Married	Dependents	Education	Self_Employed	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History	Property_Area
0	LP001015	Male	Yes	0	Graduate	No	5720	0	110.0	360.0	1.0	Urban
1	LP001022	Male	Yes	1	Graduate	No	3076	1500	126.0	360.0	1.0	Urban
2	LP001031	Male	Yes	2	Graduate	No	5000	1800	208.0	360.0	1.0	Urban
3	LP001035	Male	Yes	2	Graduate	No	2340	2546	100.0	360.0	NaN	Urban
4	LP001051	Male	No	0	Not Graduate	No	3276	0	78.0	360.0	1.0	Urban

1. The dataset has been imported from Google Drive.
2. As we begin our Exploratory Data Analysis (EDA), I've named the dataset 'df'.
3. df.head(): By default, displays the top 5 rows of the dataframe(Dataset).
4. The dataset comprises of 367 rows and 12 columns.

```
[149] df.shape
      rows=df.shape[0]
      cols=df.shape[1]
      print(f"The dataset has {rows} rows and {cols} columns")
```

→ The dataset has 367 rows and 12 columns

Step 3: Basic Data Inspection

```
[159] #Summarize basic statistics (mean, median, standard deviation, etc.) for the numeric columns  
df.describe()
```

	ApplicantIncome	CoapplicantIncome	LoanAmount	Loan_Amount_Term	Credit_History
count	367.000000	367.000000	362.000000	361.000000	338.000000
mean	4805.599455	1569.577657	136.132597	342.537396	0.825444
std	4910.685399	2334.232099	61.366652	65.156643	0.380150
min	0.000000	0.000000	28.000000	6.000000	0.000000
25%	2864.000000	0.000000	100.250000	360.000000	1.000000
50%	3786.000000	1025.000000	125.000000	360.000000	1.000000
75%	5060.000000	2430.500000	158.000000	360.000000	1.000000
max	72529.000000	24000.000000	550.000000	480.000000	1.000000

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 367 entries, 0 to 366  
Data columns (total 12 columns):  
#   Column                Non-Null Count  Dtype  
---  ---  
0   Loan_ID                367 non-null   object  
1   Gender                 356 non-null   object  
2   Married                 367 non-null   object  
3   Dependents              357 non-null   object  
4   Education               367 non-null   object  
5   Self_Employed           344 non-null   object  
6   ApplicantIncome         367 non-null   int64  
7   CoapplicantIncome       367 non-null   int64  
8   LoanAmount              362 non-null   float64  
9   Loan_Amount_Term        361 non-null   float64  
10  Credit_History           338 non-null   float64  
11  Property_Area            367 non-null   object  
dtypes: float64(3), int64(2), object(7)  
memory usage: 34.5+ KB
```

1. `df.describe()`: It gives us summary statistic of data set, by default return for only numerical columns and if we want for all columns including categorical, we can: `Data . describe (include="all")` Or can specify datatypes of preferred columns we want.

2. `df.info()`: Displays Information of the dataframe indicating null values and data types of columns.

3. There is a huge difference between 75% quantile and Max value in two columns : Applicant Income Co-Applicant Income which signifies presence of outliers.

Step 4: Data Cleaning and Pre Processing

1. Dropping Duplicated values if any.

```
[160] df.duplicated().sum()
```

```
np.int64(0)
```

```
[161] df.drop_duplicates(inplace=True)
```

```
[162] df.shape
```

```
(367, 12)
```

Step 4: Data Cleaning and Pre Processing

2. Handling Missing Values

	0
Loan_ID	0
Gender	11
Married	0
Dependents	10
Education	0
Self_Employed	23
ApplicantIncome	0
CoapplicantIncome	0
LoanAmount	5
Loan_Amount_Term	6
Credit_History	29
Property_Area	0

✓ Handling missing values - we will check the percentage of missing values:-

- if missing values are greater than 25% we will drop the column (default you are domain expert)
- if less than 25% values are missing then we will cap the values using statistical method

```
[165] #percentage  
(df.isnull().sum()/len(df))*100
```

	0
Loan_ID	0.000000
Gender	2.997275
Married	0.000000
Dependents	2.724796
Education	0.000000
Self_Employed	6.267030
ApplicantIncome	0.000000
CoapplicantIncome	0.000000
LoanAmount	1.362398
Loan_Amount_Term	1.634877
Credit_History	7.901907
Property_Area	0.000000

dtype: float64

Step 4: Data Cleaning and Pre Processing

2. Handling Missing Values

There are total 84 Null values present in our dataset. Since, all are less than 25% hence have to deal with missing values by replacing categorical with mode and numerical with either mean or median depending on presence of Outliers.

```
✓ [166] list_of_cats=['Gender','Self_Employed']  
0s     for i in list_of_cats:  
       df[i]=df[i].fillna(df[i].mode()[0])
```

	0
Loan_ID	0
Gender	0
Married	0
Dependents	10
Education	0
Self_Employed	0
ApplicantIncome	0
CoapplicantIncome	0
LoanAmount	5
Loan_Amount_Term	6
Credit_History	29
Property_Area	0

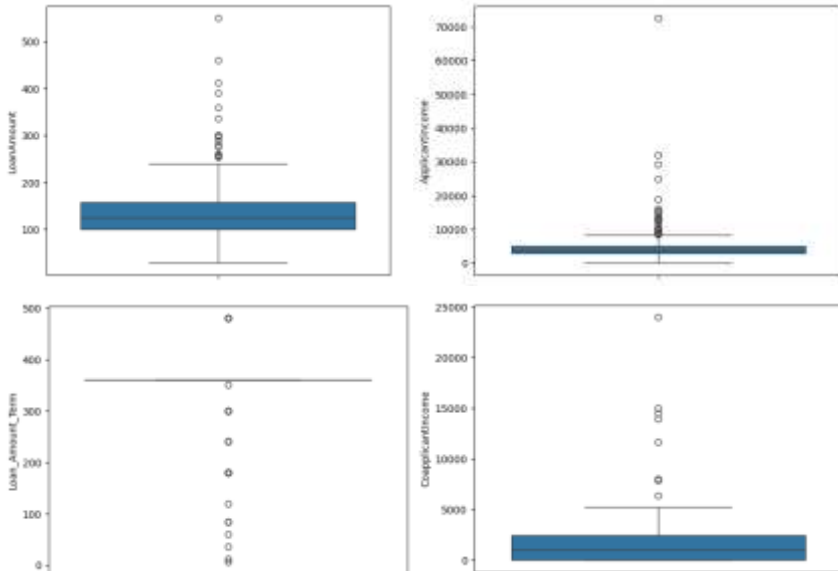
dtype: int64

```
[170] #There are outliers, hence filling the missing values with median  
      list_of_nums=['LoanAmount','Loan_Amount_Term','Credit_History']  
      for i in list_of_nums:  
        df[i]=df[i].fillna(df[i].median())
```

	0
Loan_ID	0
Gender	0
Married	0
Dependents	0
Education	0
Self_Employed	0
ApplicantIncome	0
CoapplicantIncome	0
LoanAmount	0
Loan_Amount_Term	0
Credit_History	0
Property_Area	0

Step 4: Data Cleaning and Pre Processing

3. Handling Outliers : Outliers are extreme values in a dataset that deviate significantly from the majority of the data. After generating a box plot for the 'sale-price', we identified the presence of outliers in this column .Hence, storing all the numerical columns in a variable and then handling outliers together using Inter Quartile Range method. Setting upper bound and lower bound



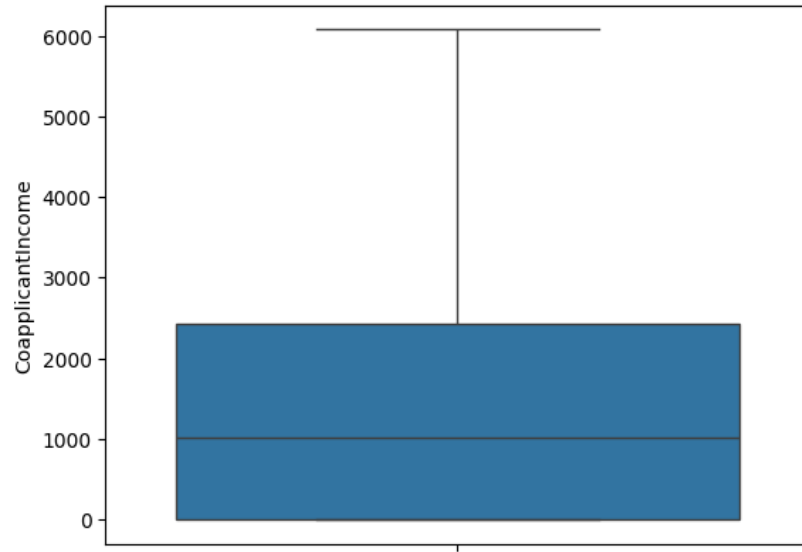
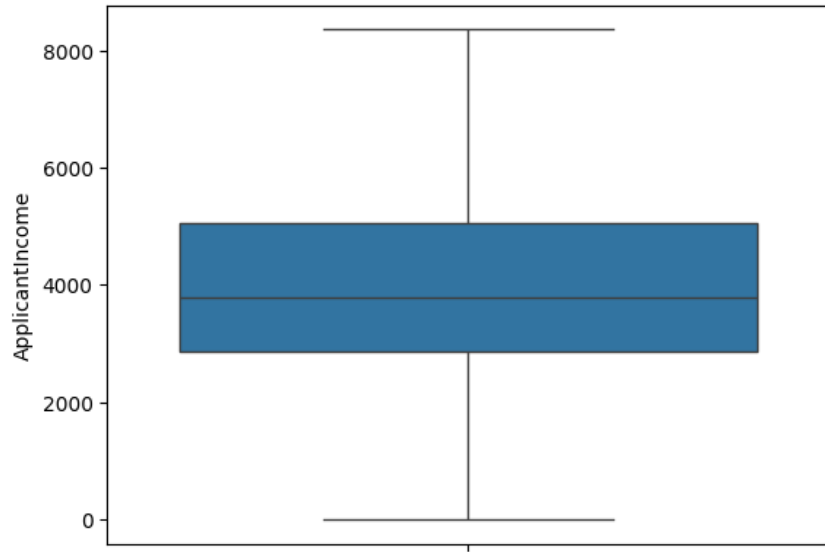
```
## Handling outliers: IQR method
list_of_nums=['LoanAmount','Loan_Amount_Term','Credit_History']
for i in list_of_nums:
    q1 = df[i].quantile(0.25)
    q3 = df[i].quantile(0.75)
    iqr = q3 - q1
    print(f"Q1:{q1}")
    print(f"Q3:{q3}")
    print(f"IQR:{iqr}")
```

```
[ ] ## formulate UL and lower LL
    UL = q3+1.5*iqr
    LL = q1-1.5*iqr
    df[i] = np.where(df[i]>UL,UL,
                     np.where(df[i]<LL,LL,
                              df[i]))
```

Step 4: Data Cleaning and Pre Processing

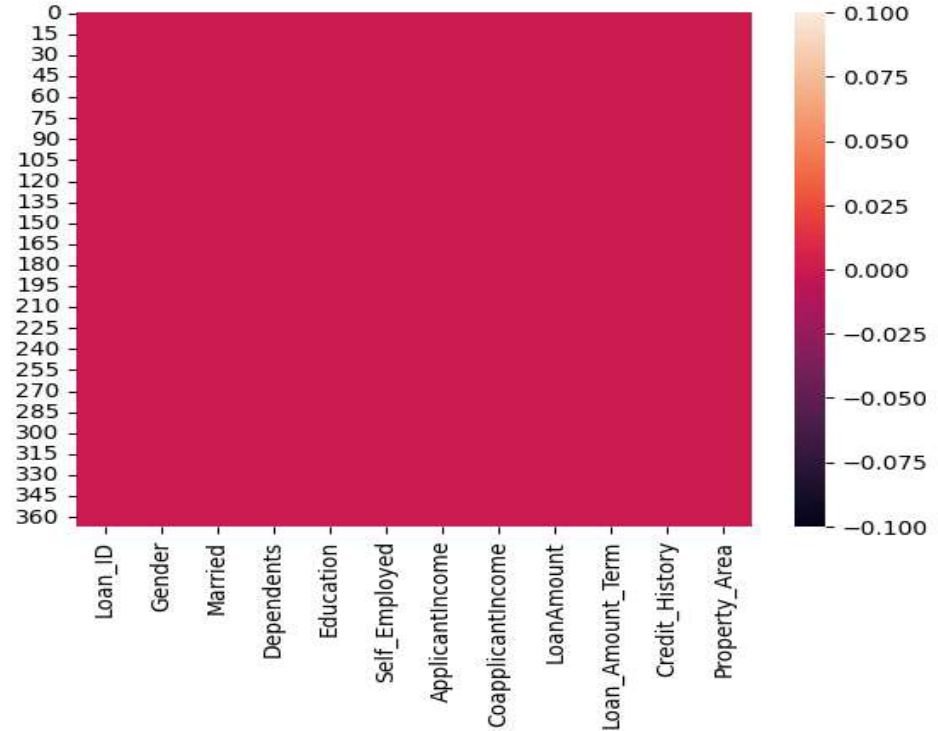
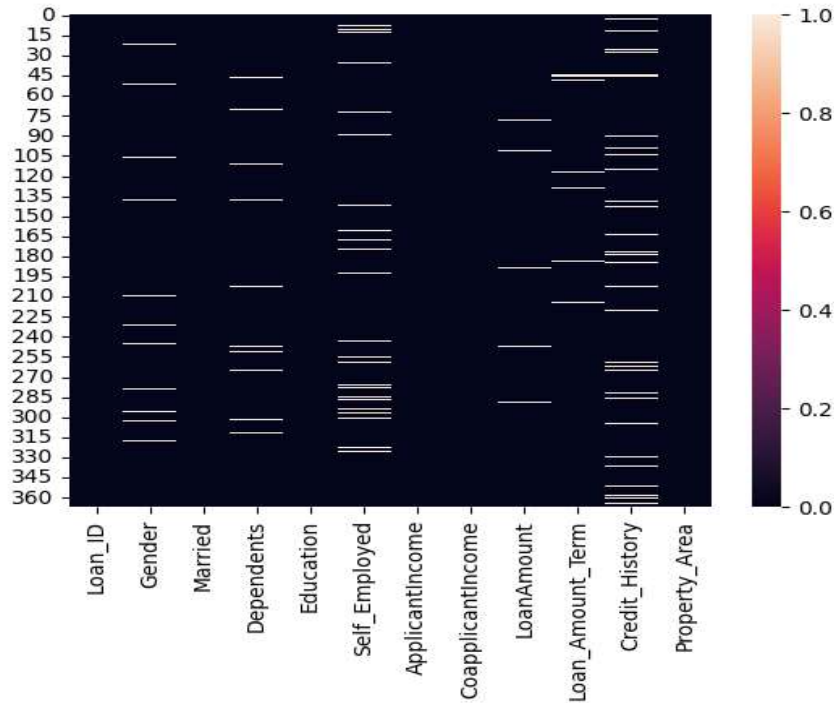
Applicant Income : This feature did contain a few Outliers which has been handled using the Inter-Quartile Range (IQR) Method.As, few applicant had 0 income, which is practically not possible in Home loan.

Co-Applicant Income: Outliers had been handled here too with Inter-Quartile Range (IQR) Method.



Step 4: Data Cleaning and Pre Processing

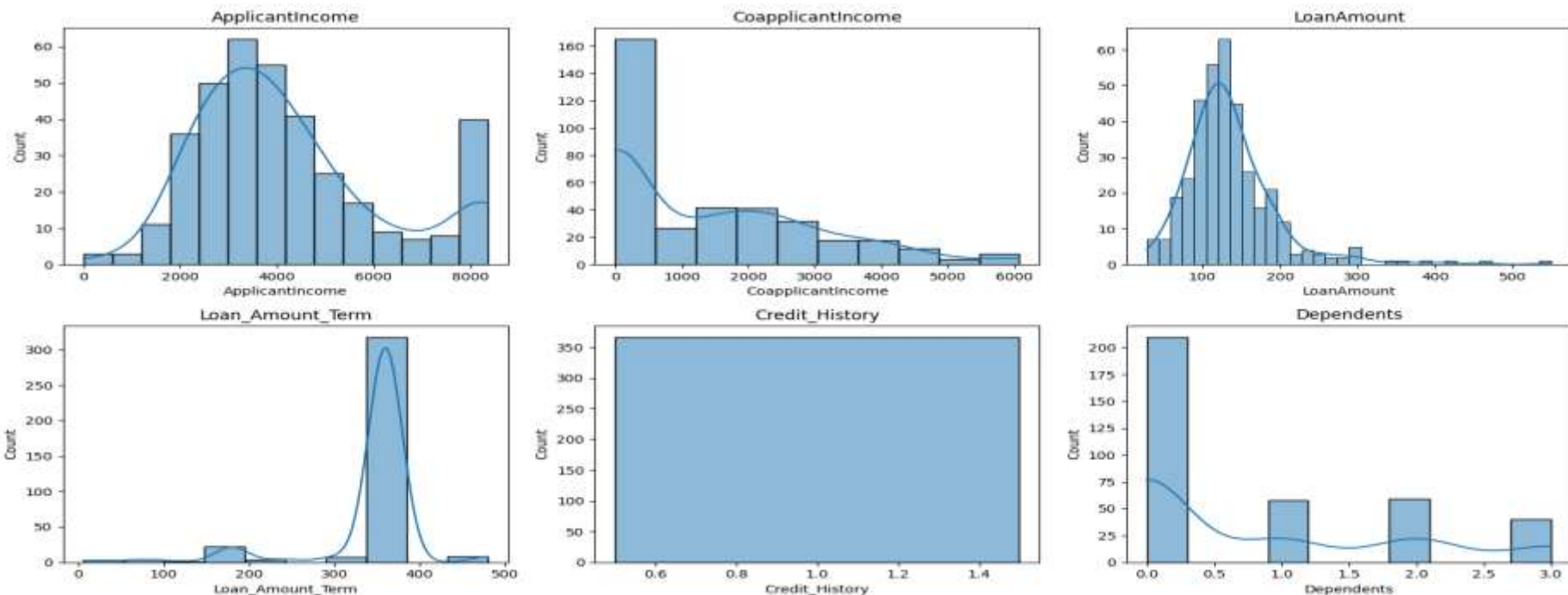
Before Vs After Handling Missing Values



Step 5: Exploratory Data Analysis and Visualization

1. Univariate Analysis

Histograms: Plot the frequency distribution of key numeric variables



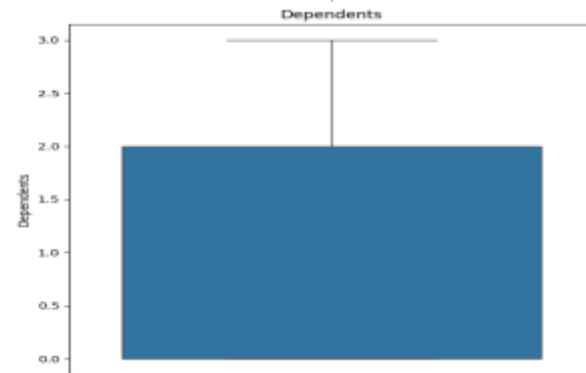
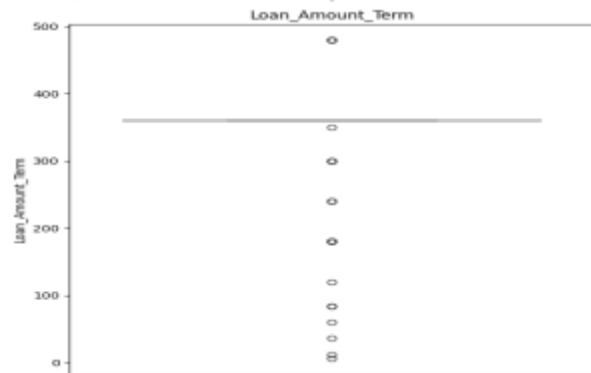
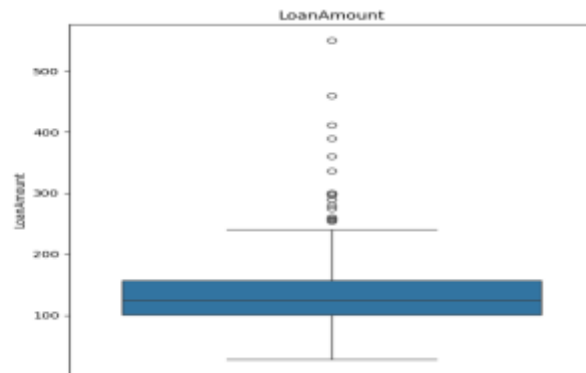
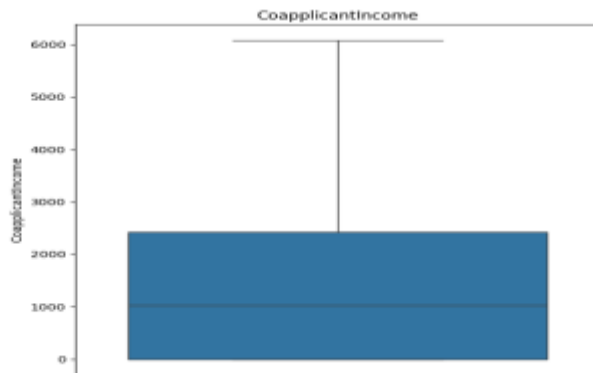
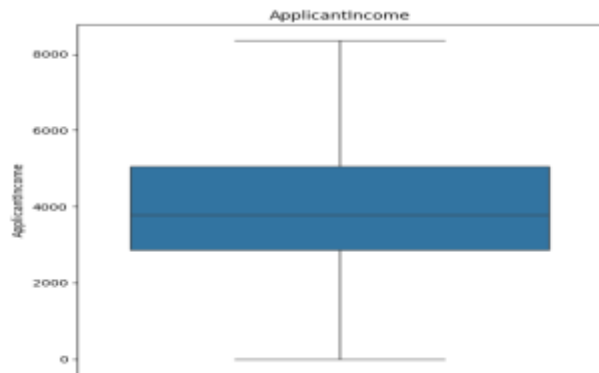
Step 5: Exploratory Data Analysis and Visualization

Findings:

1. **Applicant Income** The distribution is right-skewed, meaning most applicants have relatively low incomes. This suggests that a large portion of loan seekers fall into a lower-income bracket, which can influence loan amounts and approval chances.
2. **Co-applicant Income** Like the applicant income, this distribution is heavily right-skewed. Most co-applicants earn less than ₹5,000, indicating limited financial contribution from secondary applicants in many cases.
3. **Loan Amount** The loan amount distribution is moderately right-skewed. A large number of applicants request loans in the ₹100–₹200K range, highlighting the typical affordability level and common loan expectations.
4. **Loan Amount Term** The majority of loan terms are clustered around 360 months (30 years). This implies that most applicants prefer long-term repayment plans, possibly for home purchases or larger investments.
5. **Credit History** The histogram shows a clear divide between applicants with and without a valid credit history (values mostly 1 or 0). A good credit history (1) is far more frequent, reaffirming its importance in loan approvals.
6. **Dependents** Most applicants report having 0 to 2 dependents, with '3+' being less common. This reflects the typical family size of applicants, which can affect financial obligations and loan eligibility.

Step 5: Exploratory Data Analysis and Visualization

Box Plots: Identify potential outliers and visualize the spread of data.



Step 5: Exploratory Data Analysis and Visualization

Findings:

ApplicantIncome: The distribution is right-skewed with most applicants having lower incomes. Several high-income outliers are present, indicating a few applicants earn significantly more than the average.

CoapplicantIncome: A large number of entries are around zero, suggesting many applicants have no co-applicant or co-applicants without income. Outliers are evident, with some co-applicants earning substantially higher than the typical range.

LoanAmount: The loan amounts are mostly concentrated between 100 to 250 (likely in thousands), showing common loan preferences. Clear presence of outliers above 300–400, indicating a minority requesting much larger loans.

Loan_Amount_Term: The most frequent term is 360 months, showing a preference for long-term loans (likely mortgages). A few outliers exist at very short or extremely long durations, which are less common among applicants.

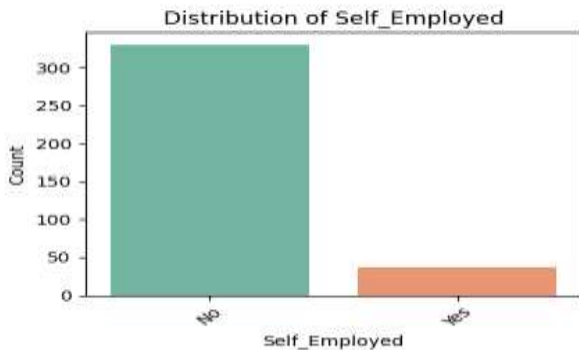
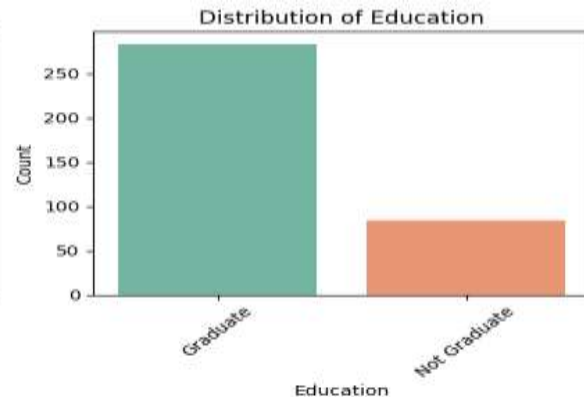
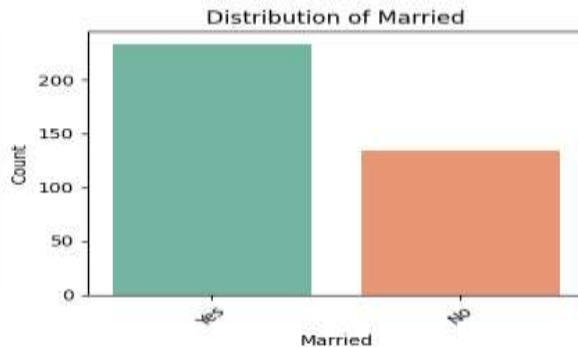
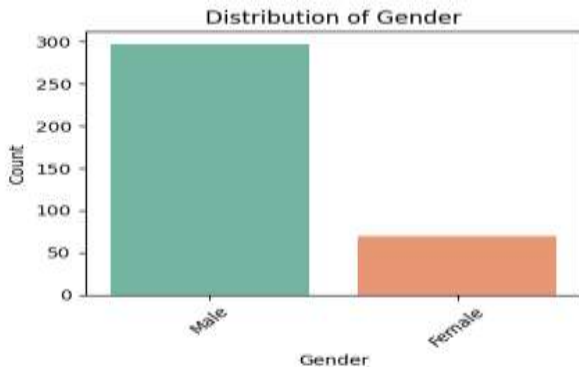
Credit_History: Very limited variation; most applicants have a credit history value of 1.

This indicates that the majority are considered creditworthy, with no significant outliers visible.

Dependents: Most applicants report 0 to 2 dependents. Values like "3+" converted to 3 do not indicate outliers, but suggest a moderately increasing family responsibility.

Step 5: Exploratory Data Analysis and Visualization

Bar Charts: Visualize the frequency distribution of categorical variables.



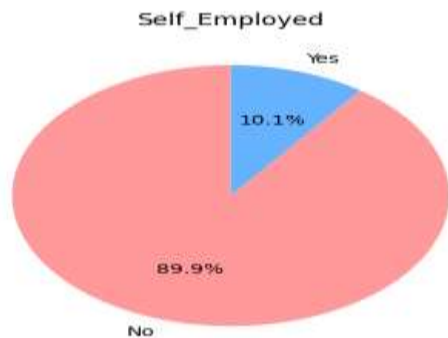
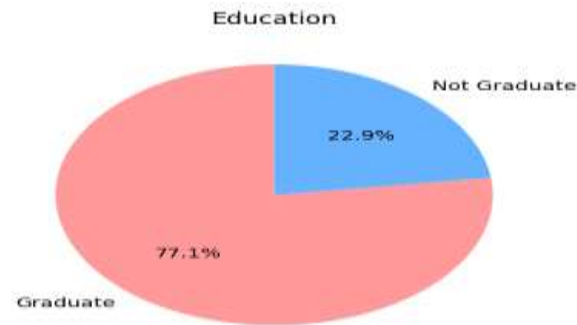
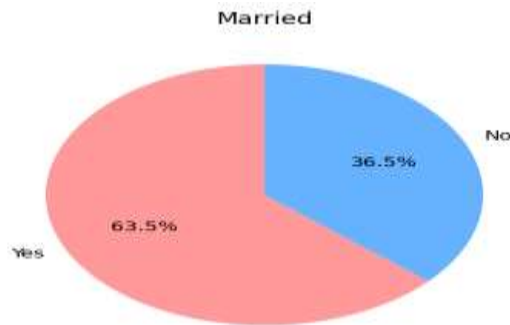
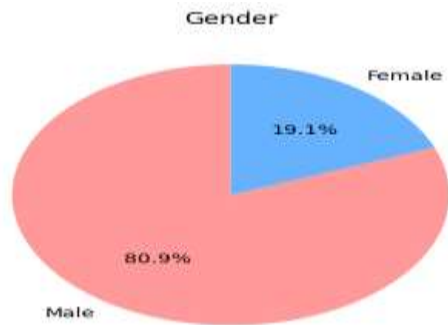
Step 5: Exploratory Data Analysis and Visualization

Findings:

1. Gender: Most loan applicants are male, suggesting either a gender-based application preference or greater financial responsibility taken by males in households.
2. Married: A large proportion of applicants are married, indicating that married individuals may have higher loan demands—possibly due to shared responsibilities or long-term financial planning.
3. Education: The majority of loan applicants are graduates. This may imply that higher education correlates with greater financial literacy or access to loan opportunities.
4. Self-Employed: Most applicants are not self-employed. This suggests that salaried individuals are more likely to apply for loans, possibly due to more stable and verifiable incomes.
5. Property Area: Applications are fairly well distributed across urban, semi-urban, and rural areas, with a slightly higher share in semi-urban regions. This reflects widespread housing needs across all area types.

Step 5: Exploratory Data Analysis and Visualization

Pie Charts: Represent the composition of categorical variables



Step 5: Exploratory Data Analysis and Visualization

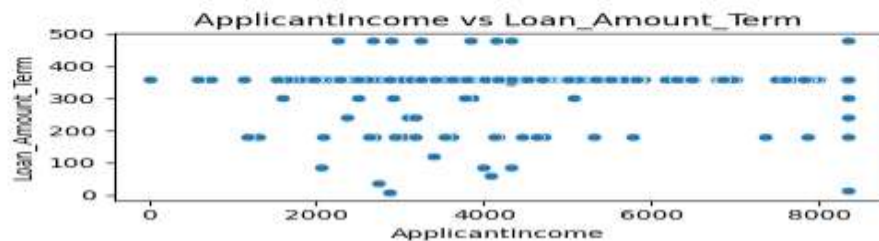
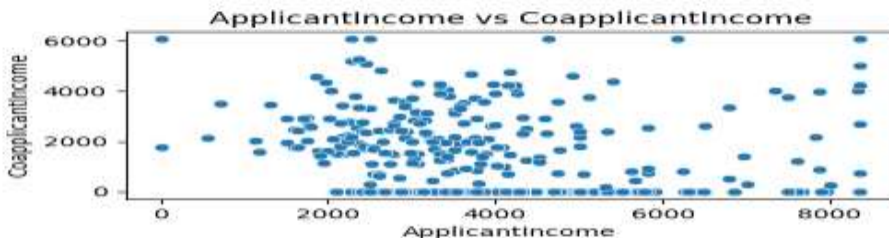
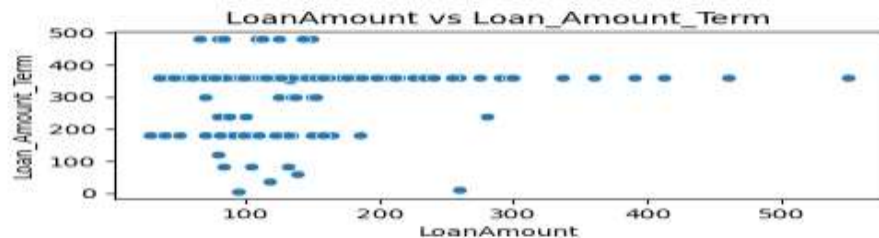
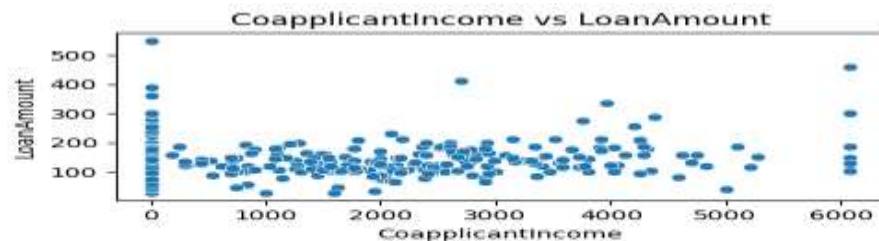
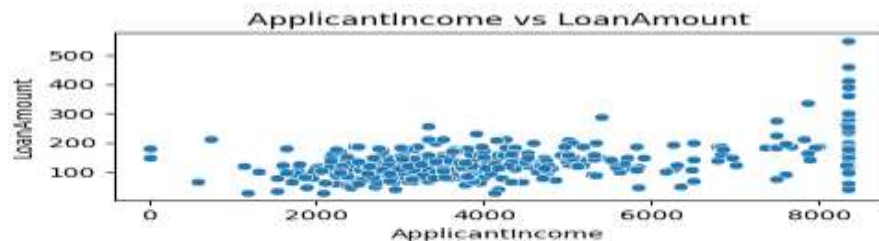
Findings:

1. Gender Distribution: Around 81%(80.9%) of the loan applicants are male, indicating a significant gender imbalance in loan applications.
2. Marital Status Distribution: Approximately 63.5% of the applicants are married, suggesting that married individuals may be more likely to apply for loan due to shared responsibilities.
3. Education Distribution: A majority (around 77%) of the applicants are graduates, indicating a higher likelihood of loan applications from individuals with higher education levels, also people not graduated yet, may likely have less financial or family burden.
4. Self-Employed Distribution: Only about 10% of the applicants are self-employed, while about 89% employed, which implies that a majority of loan applications come from salaried individuals also more chances of loan repay at stipulated tenure.
5. Property Area Distribution: The distribution of applicants across property areas (Semiurban, Urban, Rural) is relatively balanced, with Urban and Semiurban areas have a competitive equivalent demand, and Urban with slightly higher.

Step 5: Exploratory Data Analysis and Visualization

2.2 Bivariate Analysis

Create scatter plots to explore relationships between pairs of numeric variables.



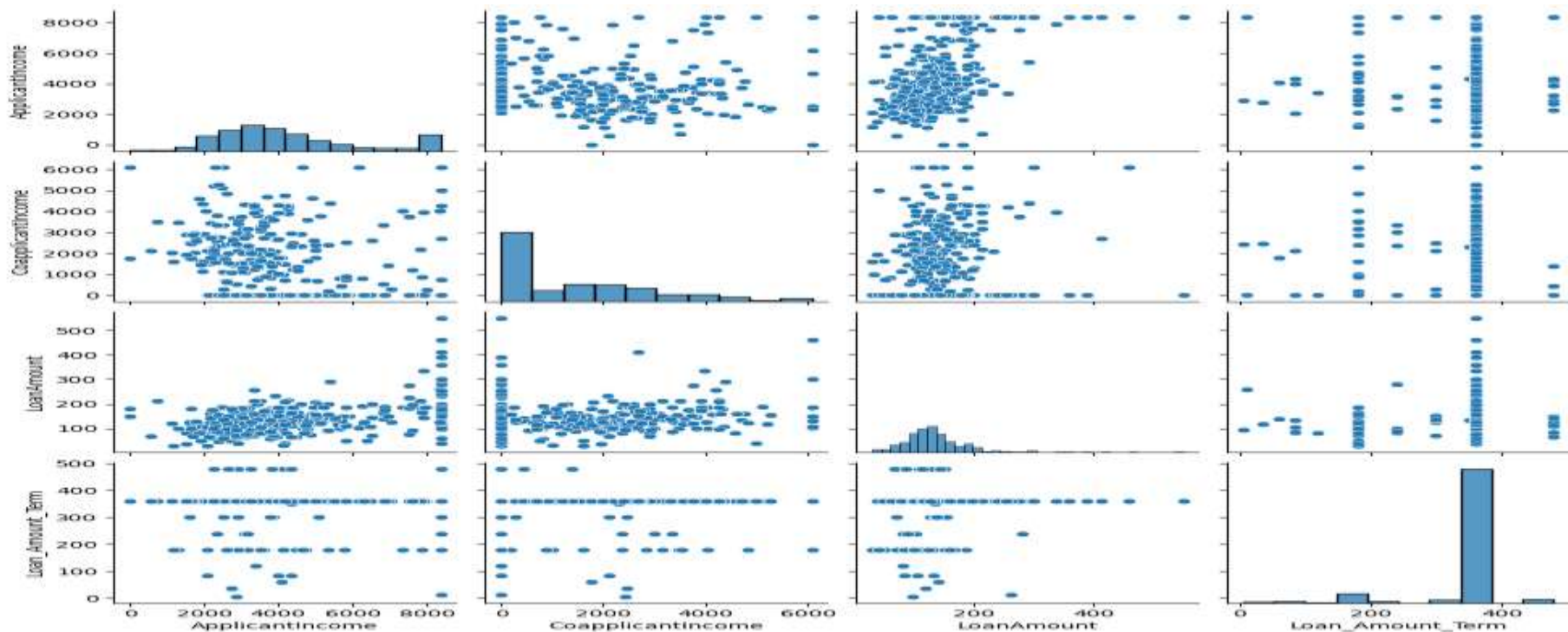
Step 5: Exploratory Data Analysis and Visualization

Findings:

1. ApplicantIncome vs LoanAmount: Weak positive trend observed where igher applicant income slightly corresponds to higher loan amounts which means some high-income applicants still request small loans.
2. CoapplicantIncome vs LoanAmount:No strong visible correlation. Many co-applicants have zero income, suggesting they may not influence loan amount significantly.
3. LoanAmount vs Loan_Amount_Term: Most loan terms cluster around 360 months regardless of the loan amount.
4. ApplicantIncome vs CoapplicantIncome: Majority of applicants have either high applicant income or co-applicant income, not both.
5. ApplicantIncome vs Loan_Amount_Term: No clear relationship.

Step 5: Exploratory Data Analysis and Visualization

Use pair plots (scatter matrix) to visualize interactions between multiple numeric variables simultaneously.



Step 5: Exploratory Data Analysis and Visualization

Findings:

1. Applicant Income vs Loan Amount: A positive correlation exists, meaning that as 'Applicant Income' rises, the 'Loan Amount' they qualify for also tends to increase.
2. Co-applicant Income vs Loan Amount: No significant correlation is observed. The 'Loan Amount' does not appear to be strongly influenced by 'Co-applicant Income'.
3. Loan Amount vs Loan Amount Term: No clear correlation is found. The loan duration does not seem to have a strong relationship with the loan amount.

Step 5: Exploratory Data Analysis and Visualization

Investigate the relationship between categorical and numeric variables using box plots or violin plots

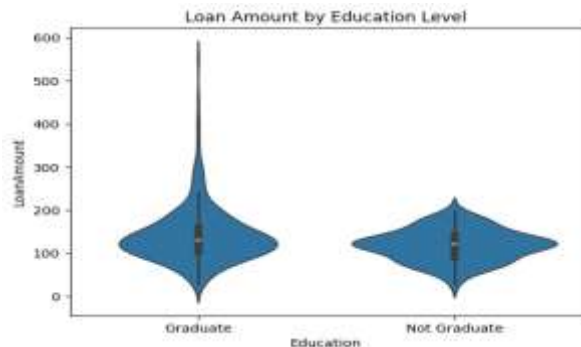


Chart 1 :

Graduates tend to apply for a wider range of loan amounts, including higher ones.

Not Graduates mostly apply for moderate loans, with fewer extreme values.

Education level may influence financial confidence or eligibility.

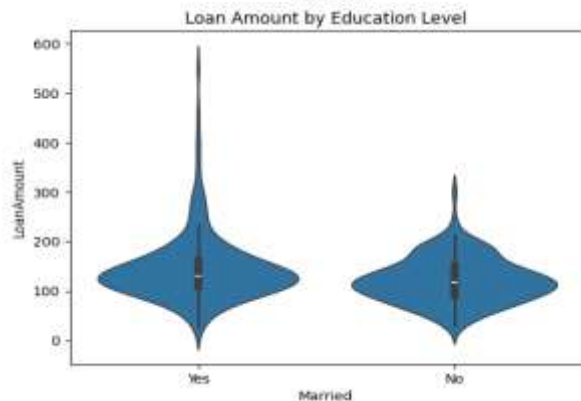


Chart 2:

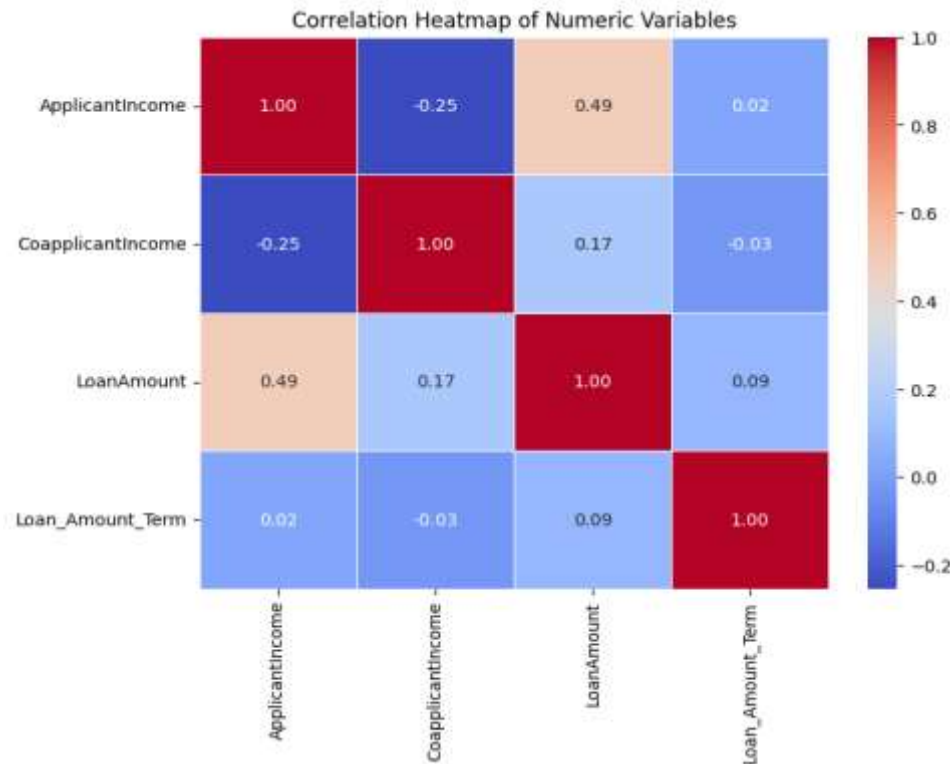
Married applicants show a broader spread, suggesting they may apply for both higher and lower loan amounts.

Unmarried applicants mostly stick to mid-range loans.

Marital status could indirectly reflect joint financial planning, allowing for larger loan requests.

Step 5: Exploratory Data Analysis and Visualization

2.3 Multivariate Analysis



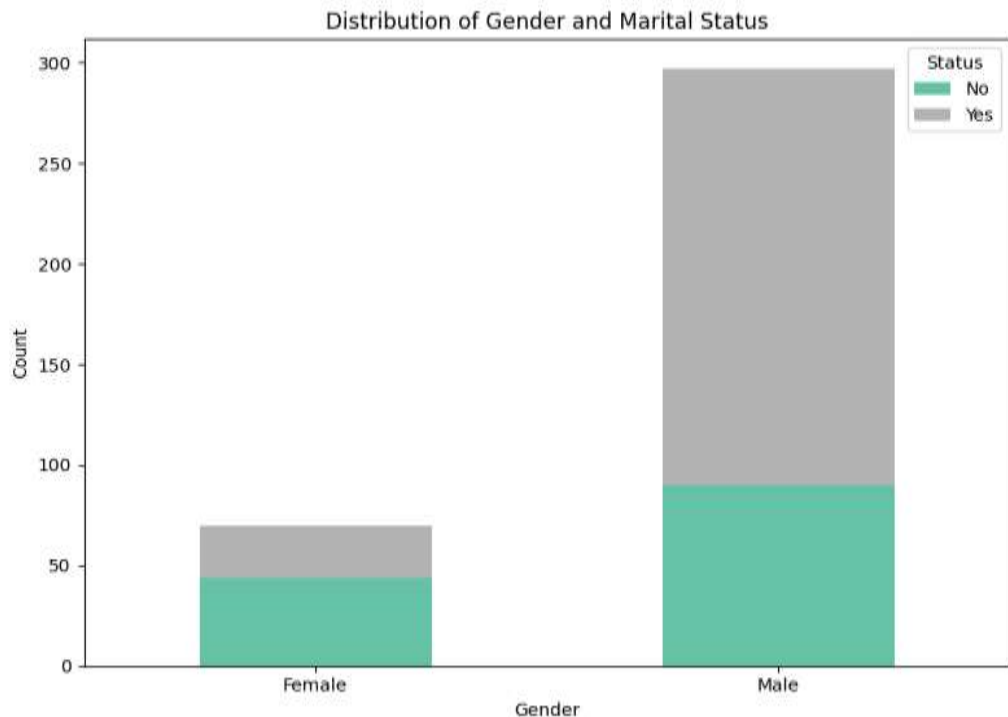
ApplicantIncome – LoanAmount 0.49 Moderate positive correlation—higher income often leads to higher loan amount.

CoapplicantIncome – LoanAmount 0.17 Weak correlation—coapplicant income has limited impact on loan amount.

LoanAmount – Loan_Amount_Term 0.09 No relationship almost 0—term length doesn't influence loan amount. Applicant vs Coapplicant Income -0.25 Mild negative trend—if one earns more, the other tends to earn less (often 0).

Step 5: Exploratory Data Analysis and Visualization

Create a stacked bar chart to show the distribution of categorical variables across multiple categories



Males form the majority in the dataset.

Among males, the "Yes" (likely married) group is significantly larger than "No".

Females are fewer in number, and their marital status is more balanced, though slightly skewed towards "No".

Step 6: Key Insights

1. Income Distribution and Impact

- **Applicant Income** is **right-skewed**, indicating that most applicants fall into a lower-income category. This suggests that many loan seekers may require smaller loan amounts and have limited repayment capacity.
- A few **high-income outliers** exist, but these are rare and likely represent a small subset of premium customers.
- **Co-applicant Income** also shows a right-skewed distribution with many entries at or near zero, meaning many applications either don't have a co-applicant or the co-applicant has no reported income.

2. Credit History Significance

- A significant majority of applicants have a **credit history score of 1**, showing that most are considered creditworthy.
- This feature is likely a **critical driver** for loan approval decisions and a key indicator of applicant reliability.

Step 6: Key Insights

3. Loan Amount and Terms

- Most **loan amounts** requested fall between ₹100,000–₹250,000, showing common affordability and expectations.
- A **moderate positive correlation (0.49)** exists between **applicant income and loan amount**, reinforcing that higher earners qualify for larger loans.
- The **loan term** is usually **360 months**, indicating a dominant preference for long-term loans—likely for housing.
- **Loan amount and loan term show no significant correlation**, implying the duration is selected independently of the amount.

4. Geographic and Socioeconomic Factors

- Loan applications are fairly distributed across **urban, semi-urban, and rural** areas, though **semi-urban** regions slightly lead. Urban areas still represent the highest share.
- **Property area** does not show extreme imbalance, suggesting fairly even loan demand across regions.

Step 6: Key Insights

5. Demographic Patterns

- **Gender Imbalance:** Approximately **81%** of applicants are **male**, indicating a stark gender disparity in financial decision-making or accessibility.
- **Marital Status:** Around **63.5%** are **married**, suggesting a strong correlation between marriage and financial planning needs like loans.
- **Education Level:** **77%** of applicants are **graduates**, implying that education may contribute to financial literacy and access to credit.
- **Employment Type:** Only **10%** are self-employed; salaried individuals dominate the applicant pool, reflecting banks' preference for stable income sources.
- **Dependents:** Most applicants report **0–2 dependents**, indicating moderate financial responsibility levels among loan seekers.

Step 6: Key Insights

6. Correlations and Trends

- **Weak correlation (0.17)** between **co-applicant income and loan amount** implies that co-applicant income doesn't strongly influence loan decisions.
- **Mild negative correlation (-0.25)** between applicant and co-applicant income hints that when one earns more, the other often earns less or is unemployed.
- **Graduates and married applicants** tend to request **wider ranges** of loan amounts, including higher values, possibly due to shared income or better understanding of loan mechanisms.

Recommendations

1. Most applicants fall in the lower income range, banks should focus on affordable EMI structures, low-interest schemes, and flexible repayment options.
2. Close the gender gap by offering customized financial programs for women. Support self-employed individuals with documentation assistance, alternative credit scoring models, and business-friendly loan products.
3. Since educated individuals are more likely to apply for and manage loans effectively, tailor premium loan products and long-term investment loans for this group.
4. Educate applicants that a longer term doesn't necessarily increase loan approval odds or reduce risk, especially since no strong relationship exists between term length and loan amount.
5. Encourage applicants to include co-applicants with income, where possible, to increase total income and strengthen eligibility.





**Thankyou.
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